

Formalization of a Proof Calculus for Incremental Linearization for NTA Satisfiability

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¹Equal contribution

Nonlinear Arithmetic and Transcendental Functions

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- ▷ Applications in formal verification of dynamic systems, digital signal processing and motion planning for robots
 - ▶ Automation in proof assistants, if provided the necessary tooling
- ▷ QF_NRA requires exponential time to solve and QF_NTA has been shown to be undecidable

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- ▷ Given a formula ψ in QF_NTA, a new formula ψ' is obtained by abstracting each transcendental function and variable multiplication as an uninterpreted function
- ▷ ψ' is thus a formula in QF_UFLRA. If ψ' is unsatisfiable, then so is ψ

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Example

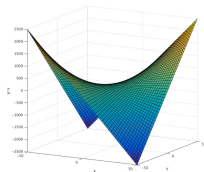
Let $\psi = x^2 + y^2 \leq 2 \wedge (x \leq -1.1 \vee x \geq 1.1) \wedge (y \leq -1.1 \vee y \geq 1.1)$. ψ' is the same, but instead of x^2 and y^2 we have $f_*(x, x)$ and $f_*(y, y)$. It is possible to interpret f_* in such a way that ψ' is satisfied, but the original formula is unsatisfiable.

Incremental Linearization

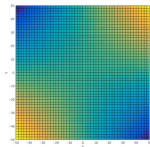
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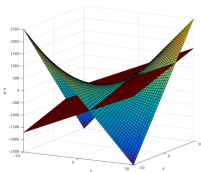
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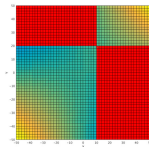
(a) $x * y$



(b) $x * y$ (top view)



(c) $x * y$ and tangent plane

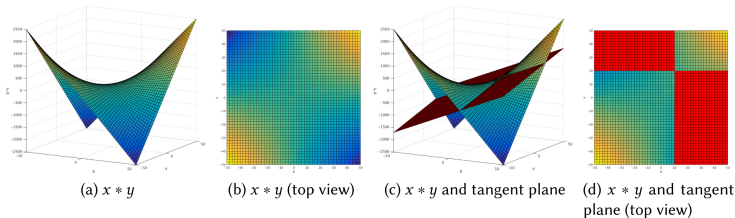


(d) $x * y$ and tangent plane (top view)

Source: Cimatti et al.

Incremental Linearization

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Source: Cimatti et al.

- ▷ Spurious solutions involving transcendental functions are blocked using their *Taylor polynomial*

Incremental Linearization

```
1: function INCREMENTALLINEARIZATION( $\psi$ )
2:    $\psi' \leftarrow \text{INITIALABSTRACTION}(\psi)$ 
3:    $\Gamma \leftarrow \emptyset$ 
4:   while INTIMELIMIT() do
5:      $\langle sat, \mu \rangle \leftarrow \text{SOLVEUFLRA}(\psi' \wedge \Gamma)$ 
6:     if not  $sat$  then return UNSAT
7:      $\langle sat', \Gamma' \rangle \leftarrow \text{CHECKREFINE}(\psi', \mu)$ 
8:     if  $sat'$  then return SAT
9:      $\Gamma \leftarrow \Gamma \cup \Gamma'$ 
```

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- ▷ The solver also defines a *proof calculus* composed of 21 inference rules capturing its reasoning
- ▷ Types of lemmas
 - ▶ Rules eliminating spurious solutions
 - ▶ Phase shifting for $\sin$
 - ▶ Bounding π

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- ▷ $\tan(x) = \frac{\sin(x)}{\cos(x)}$
- ▷ $\sec(x) = \frac{1}{\cos(x)}$
- ▷ $\log$ and inverse trigonometric functions are modeled with new variables. For instance, if the input formula has the term $\log(1 + x)$ we introduce a new variable y and add the assertion $\exp(y) = 1 + x$

Contributions

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 - ▶ *lean-smt*
- ▷ In this process, we identified some errors in the documentation, which were fixed
- ▷ Increase confidence in the proof calculus and, once cvc5 emits proofs using the calculus, in its implementation

Formalization of the Proof Calculus

Lemmas on multiplication - ARITH_MULT_TANGENT

$$\frac{- \mid a, b}{\forall x \forall y, xy \leq bx + ay - ab \leftrightarrow ((x \leq a \wedge y \geq b) \vee (x \geq a \wedge y \leq b))}$$

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Lemmas on multiplication - ARITH_MULT_TANGENT

theorem arithMulTangentLower (a b x y : $\mathbb{R}$) :
 $x * y \leq b * x + a * y - a * b \leftrightarrow ((x \leq a \wedge y \geq b) \vee (x \geq a \wedge y \leq b))$

theorem arithMulTangentUpper (a b x y : $\mathbb{R}$) :
 $x * y \geq b * x + a * y - a * b \leftrightarrow ((x \leq a \wedge y \leq b) \vee (x \geq a \wedge y \geq b))$

$$\frac{f_1 \wedge \cdots \wedge f_k \mid m}{m \bowtie 0}$$

Each f_i has the form $x_j \bowtie 0$. m is a monomial composed by powers of the x_i 's. The relation in the conclusion is defined based on the parity of the exponents and on the sign of each x_i

Lemmas on multiplication - ARITH_MULT_SIGN

```
lemma powNegOdd      :  $\forall \{k : \mathbb{N}\} \{r : \mathbb{R}\}, r < 0 \rightarrow \text{Odd } k \rightarrow r ^ k < 0$   
lemma powNegEven     :  $\forall \{k : \mathbb{N}\} \{r : \mathbb{R}\}, r < 0 \rightarrow \text{Even } k \rightarrow r ^ k > 0$   
lemma nonZeroEvenPow :  $\forall \{k : \mathbb{N}\} \{r : \mathbb{R}\}, r \neq 0 \rightarrow \text{Even } k \rightarrow r ^ k > 0$   
lemma powPos         :  $\forall \{k : \mathbb{N}\} \{r : \mathbb{R}\}, r > 0 \rightarrow r ^ k > 0$ 
```

```
lemma combineSigns1 :  $\forall \{a \ b : \mathbb{R}\}, a > 0 \rightarrow b > 0 \rightarrow b * a > 0$   
lemma combineSigns2 :  $\forall \{a \ b : \mathbb{R}\}, a > 0 \rightarrow b < 0 \rightarrow b * a < 0$   
lemma combineSigns3 :  $\forall \{a \ b : \mathbb{R}\}, a < 0 \rightarrow b > 0 \rightarrow b * a < 0$   
lemma combineSigns4 :  $\forall \{a \ b : \mathbb{R}\}, a < 0 \rightarrow b < 0 \rightarrow b * a > 0$ 
```

```
example (a b c : Real) :  
  a > 0  $\rightarrow$  b < 0  $\rightarrow$  c > 0  $\rightarrow$  a ^ 2 * b ^ 3 * c ^ 3 < 0 :=  
  by arithMulSign [a,b,c], [1,-1,1], [2,3,3]
```

Lemmas for transcendental functions - Bounding exp and sin

$$\frac{d \text{ odd} \mid d, c, t}{t \geq c \rightarrow \exp(t) \geq \text{maclaurin}(\exp, d, c)}$$

$$\frac{d \equiv 3 \bmod 4^* \wedge (0 < l \wedge u \leq \pi)^* \mid d, t, l, u}{t \in [l, u] \rightarrow \sin(t) \geq \text{secant}(\sin, l, u, t)}$$

Where `secant` is the secant line to the d -th Taylor polynomial at the points $(l, p(l))$ and $(u, p(u))$

Taylor's theorem and Lagrange's remainder

▷ These results follow from *Taylor's theorem* with the Lagrange form of the remainder:

$$\exists x' \in [x_0, x], f(x) - \left(\sum_{j=0}^n \frac{f^{(j)}(x_0)}{j!} (x - x_0)^j \right) = \frac{f^{(n+1)}(x')}{(n+1)!} (x - x_0)^{n+1}$$

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- ▷ This formula already existed in mathlib, but with the additional hypothesis that $x_0 < x$. We extended it to the general version

Lemmas for transcendental functions - Bounding exp and sin

```
theorem arithTransExpApproxBelow (d c t :  $\mathbb{N}$ ) :  
  Odd d  $\rightarrow$  t  $\geq$  c  $\rightarrow$  Real.exp t  $\geq$  taylorWithinEval Real.exp d Set.univ 0 c
```

```
theorem arithTransSineApproxBelowPos (d k :  $\mathbb{N}$ ) (hd : d = 4 * k + 3) (t l u :  $\mathbb{R}$ )  
  (ht : l  $\leq$  t  $\wedge$  t  $\leq$  u)  
  (hl : 0 < l) (hu : u  $\leq$  Real.pi) :  
  let p :  $\mathbb{R} \rightarrow \mathbb{R} :=$  taylorWithinEval Real.sin d Set.univ 0  
  sin t  $\geq$  ((p l - p u) / (l - u)) * (t - l) + p l
```

- ▷ Proof reconstruction using *lean-smt*
 - ▶ cvc5 emits a proof written in terms of these rules and the rest of its proof calculus (CPC)
 - ▶ lean-smt translates this to a Lean proof and checks it (the Lean theorem is inferred)
 - ▶ Or one can start with a Lean goal and use lean-smt to translate it to an SMT-LIB query

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Future Work

- ▷ Proof reconstruction using *lean-smt*
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 - ▶ Or one can start with a Lean goal and use lean-smt to translate it to an SMT-LIB query
- ▷ Challenges with real numbers
- ▷ Arbitrary π approximation

Thanks! Questions?